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Cubesat Orbiters and Landers Payload for Solar and Space Weather Observations (S&SWO)

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Abstract: Following recommendations from COSPAR PEX Panel on Exploration (Ehrenfreund et al 2012, Blanc et al 2025), and Horizon 2061 foresight (Blanc et al 2023, Foing et al 2023), we have developed cubesat and shoebox-size payload (CSP4X) for lunar, planetary and solar system exploration (Foing et al 2025). We are now designing an adapted 'Solar and Space Weather Observations Cubesat' S&SWOC. This includes a solar imaging telescope with vacuum UV filters covering 160 nm continuum, C IV, Ly alpha (Foing, Bonnet et al 1984, 1986, 1988, Wiik et al 1996) as well as an VUV hyperspectral imager. We also include an X-ray solar monitor as flown on SMART1 (Huovelin et al 2002), and solar particle and radiation sensors. We also include geocorona and auroral imager .

We are considering possible designs of S&SWOC in earth orbit (S&SWOC-EO) , lunar orbit (S&SWOC-LO), or on the lunar surface (S&SWOC-LS) in particular attached to landers in polar peak of quasi-eternal sunlight . We are considering possible designs of S&SWOC in earth orbit (S&SWOC-EO) , lunar orbit (S&SWOC-LO), or on the lunar surface (S&SWOC-LS) in particular attached to landers in polar peak of quasi-eternal sunlight .

We look also to develop ML techniques for on board processing and advanced analysis and modelling, as well as MHD predictive simulation tools for flares, CMEs and their space weather effects. (Schmieder et al 2013, 2015, 2020, Baratashvili et al 2022, 2025)

These could be also precursor for future cubesats in Mars orbit (S&SWOC-MO), or packages on Phobos Surface (S&SWOC-PS) . We also consider adapted cubesats in Jupiter Orbit (S&SWOC-JO), or packages on the surface of Icy Moons such as Callisto (S&SWOC-CS) to monitor space weather interactions with the Jupiter-Io-Icy Moons system.

COSPAR PEX

COSPAR recommendations 2025

Cubesat (S&SWOC) project

possible coordination with other groups

Telescope: 8 cm aperture

UV imagers: lambdas CIV, Ly alpha, cont 160, Mg II
option spectro : 200-400 nm

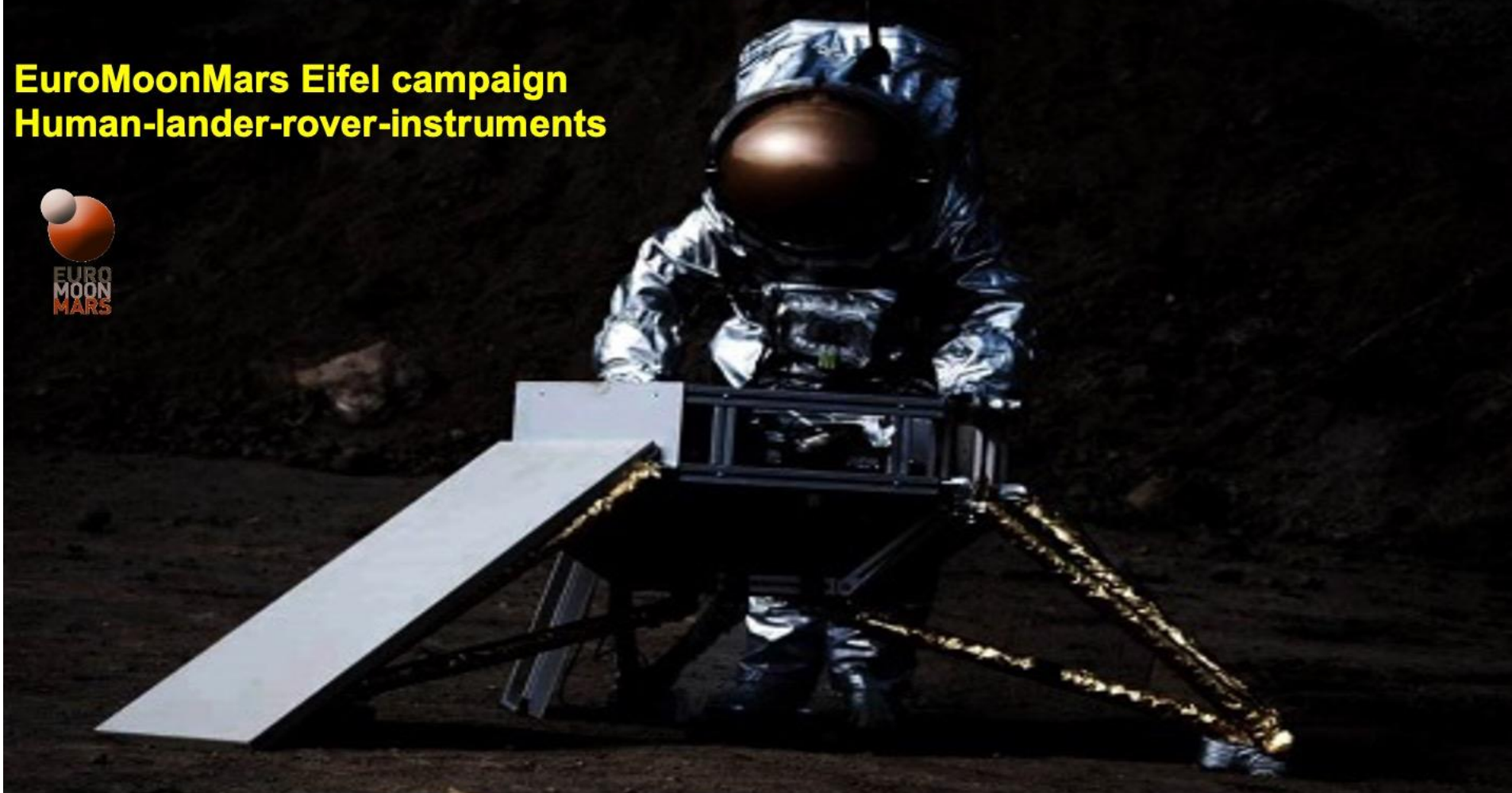
Soft X monitor (eg XSM SMART1)

Particle detector

First test on the ground with a lander

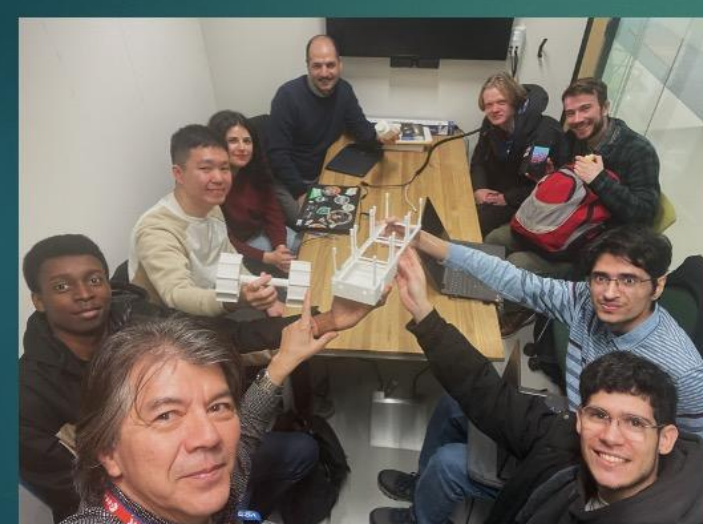


EuroMoonMars Eifel campaign
Human-lander-rover-instruments

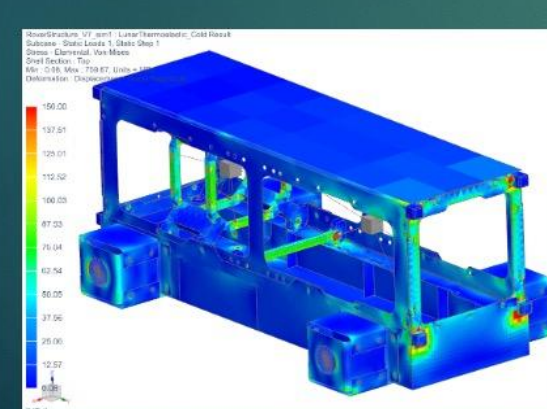
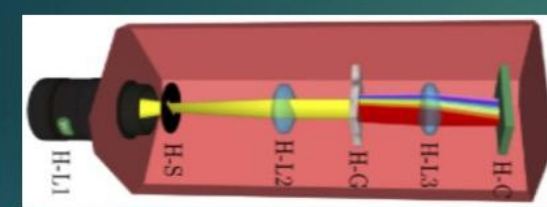


Cubesat Engineer Team

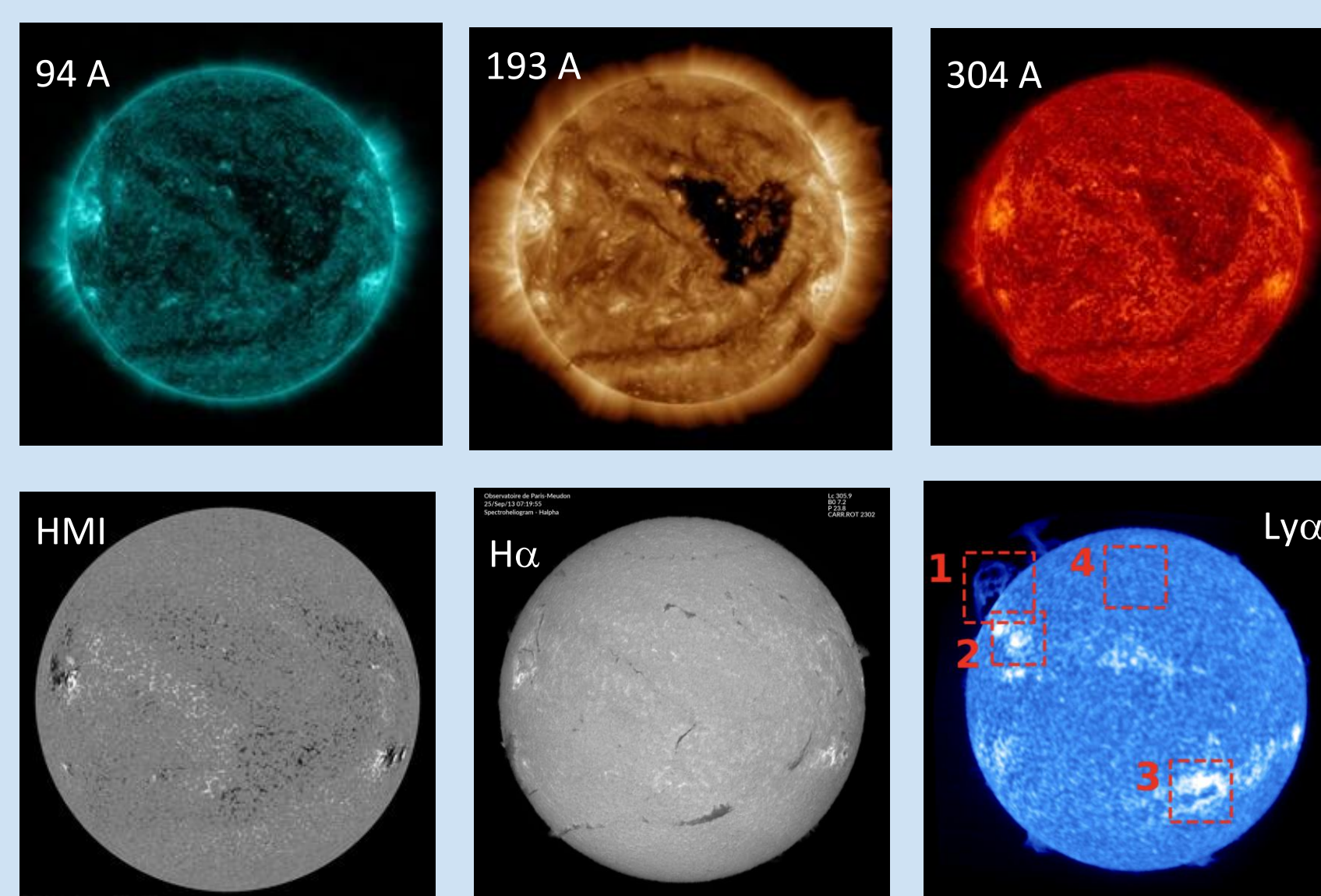
3 U cubesat breadboard



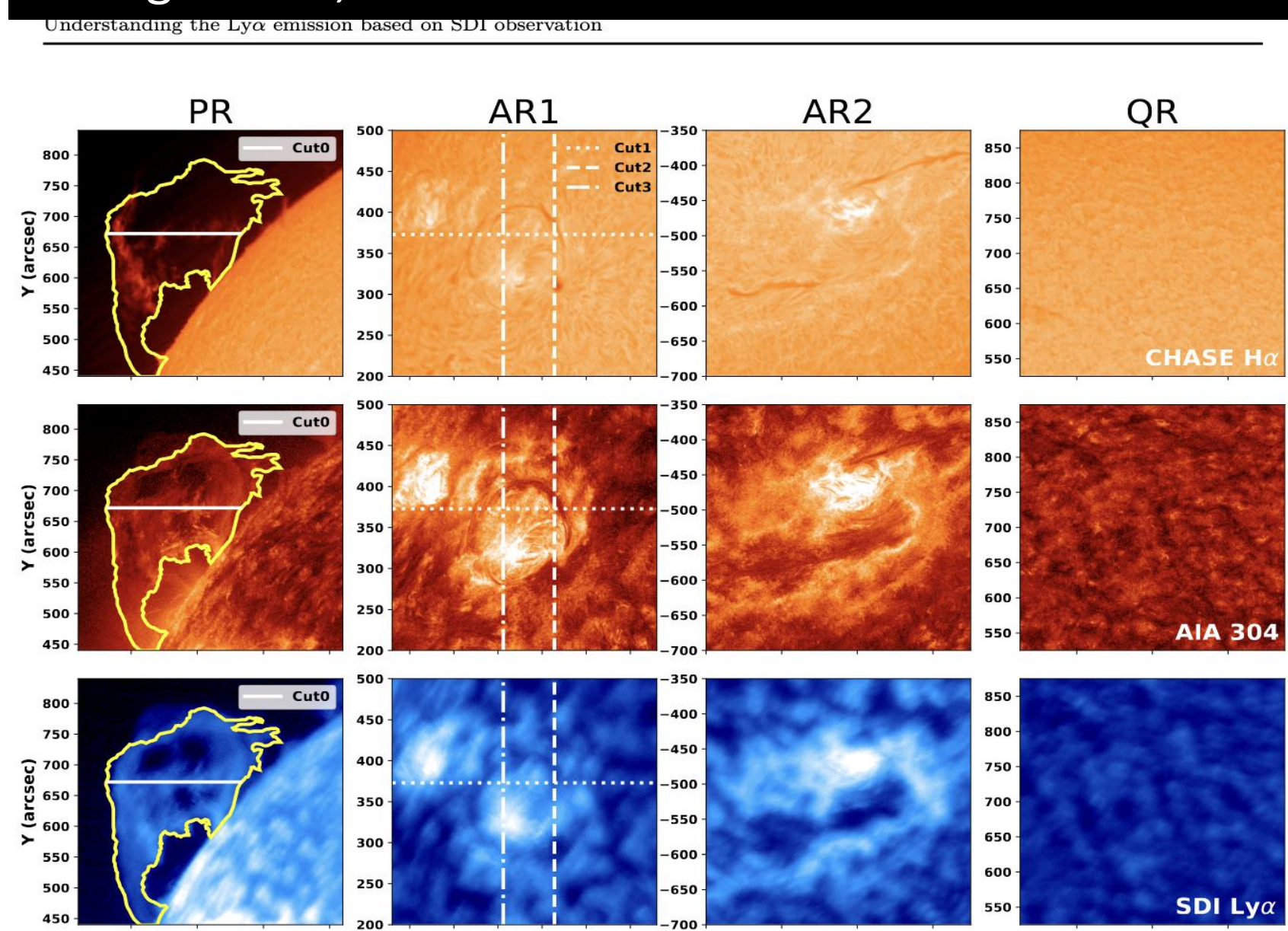
6U Cubesat concept with hyperspectral UV and XUV imager



SDO/AIA and ASO-S/SDI



Understanding the Ly α Emission Observed by the Solar Disk Imager Aboard the Advanced Space-based Solar Observatory (ASO-S –Chinese mission)
Yiliang Li et al, 2025



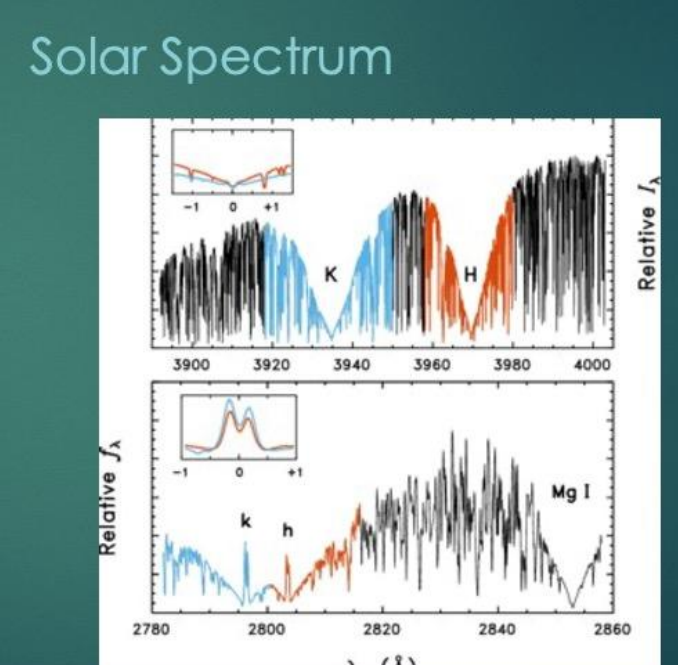
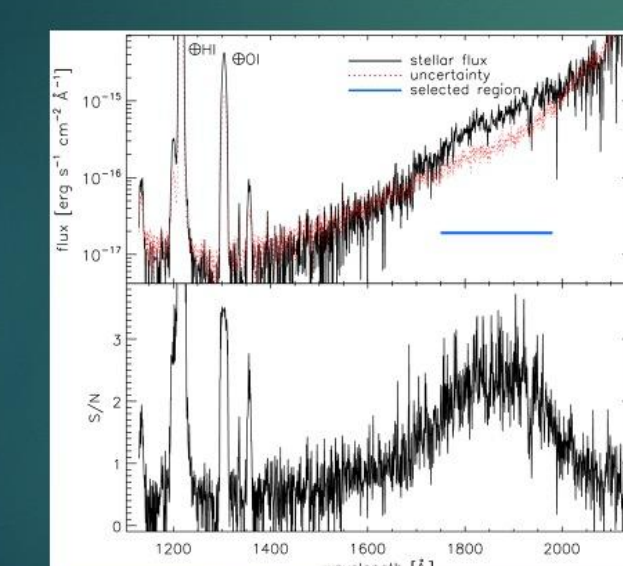
Yiliang Li et al: Solar Physics 2025, 300,78

S&SWOC

Solar and Space Weather Cubesat

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UV imaging spectrometer range



Conclusion

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References: Horizon 2061, Solar wind and Spatial Environment

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